VSI	System Requirements Document
DEPARTMENT	DOCUMENT TYPE

PROJECT NAME

VZI-F

DOCUMENT TITLE

VZI-F EMS Integration on PIS Network

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1. Acronyms and abbreviations

Acronym	Meaning
BTU	Bench Test Unit
CDRL	Contractual Deliverable Requirement List
D f E	Design for Environment
DHS	Data Handling System
ECF	Energy Calculation Function
EMI	Electromagnetic Interference
EMS	Energy Meter System
HR	Hitachi Rail STS
HV	High Voltage
DCS	Data Collecting System
LCA	Life Cycle Assessment
LRU	Last Replaceable Unit
LV	Low Voltage
MV	Medium Voltage
PDR	Preliminary Design Review
PTU	Portable Test Unit
TBD	To Be Defined
TCMS	Train Control and Monitoring System
TRS	Technical Requirements Specification
UNIFE	Association of the European Rail Industry
RAMS	Reliability, Availability, Maintainability and Safety
ICD	Interface Control Document
DFP	Design Freeze Phase
FRS	Functional Requirements Specification

Tab. 1 - Acronyms and abbreviations

2. Reference Documents

Reference	Document Number	Rev	Title
[2.1]	Capitolato tecnico prestazionale (doc. Trenitalia)	Prima emissione 22/03/2024	Servizio di Progettazione, fornitura, installazione ed omologazione del Sistema EMS sulla flotta ETR1000 I-F
[2.2]	EC010019484	01	ETR1000 VZI-R1 – Energy Meter System – Requirement Specification

Tab. 2 – Reference documents

Reference document and applicable standard indicated in document [2.2] are considered to be applicable also for this specification.

3. Introduction

The purpose of the energy metering system is to provide energy metering information or to provide energy usage data for performance analysis.

This document describes the configuration of the Energy Meter System (EMS) with reference to previous supply for VZI R1 vehicles (reference document [2.2] and the requirement for integration to the ETR1000 VZI-F fleet.

4. System description

The scope of supply of EMS include the following components (part number Hitachi of VZI R1 application are indicated):

- 2 DC voltage and current sensor (1 in car TT2, 1 in car TT7)
  [ECOMETER DC – part number HW03979552-00]
- 1 DHS on car T5
  [ECOMODULE – part number HW03979583-00]
- 2 AC voltage sensor, (1 in car T4, 1 in car T5) both connected to the DHS
  [ELETRAFO EVO – part number HW03979540-00]
- 1 Antenna GSM/GPS connected to the DHS on car T5
  [part number HW03979570-00]

In addition, the EMS receive as input 2 AC current sensor (1 in car T4, 1 in car T5), both connected to the DHS [part number EC030031224, not included in scope of supply of EMS system].

The DHS collects data from sensors on the train and perform energy calculations in both DC and AC.
The DHS and the DC sensor are connected to PIS ethernet network.

The two DC Sensors and the DHS are connected to the PIS Network and use it to exchange the necessary information for the Energy Meter System functionality.

The DHS will use both antenna and PIS network to be able to transmit data to wayside server.

The EMS system, component and connection to network is shown in figure 1.

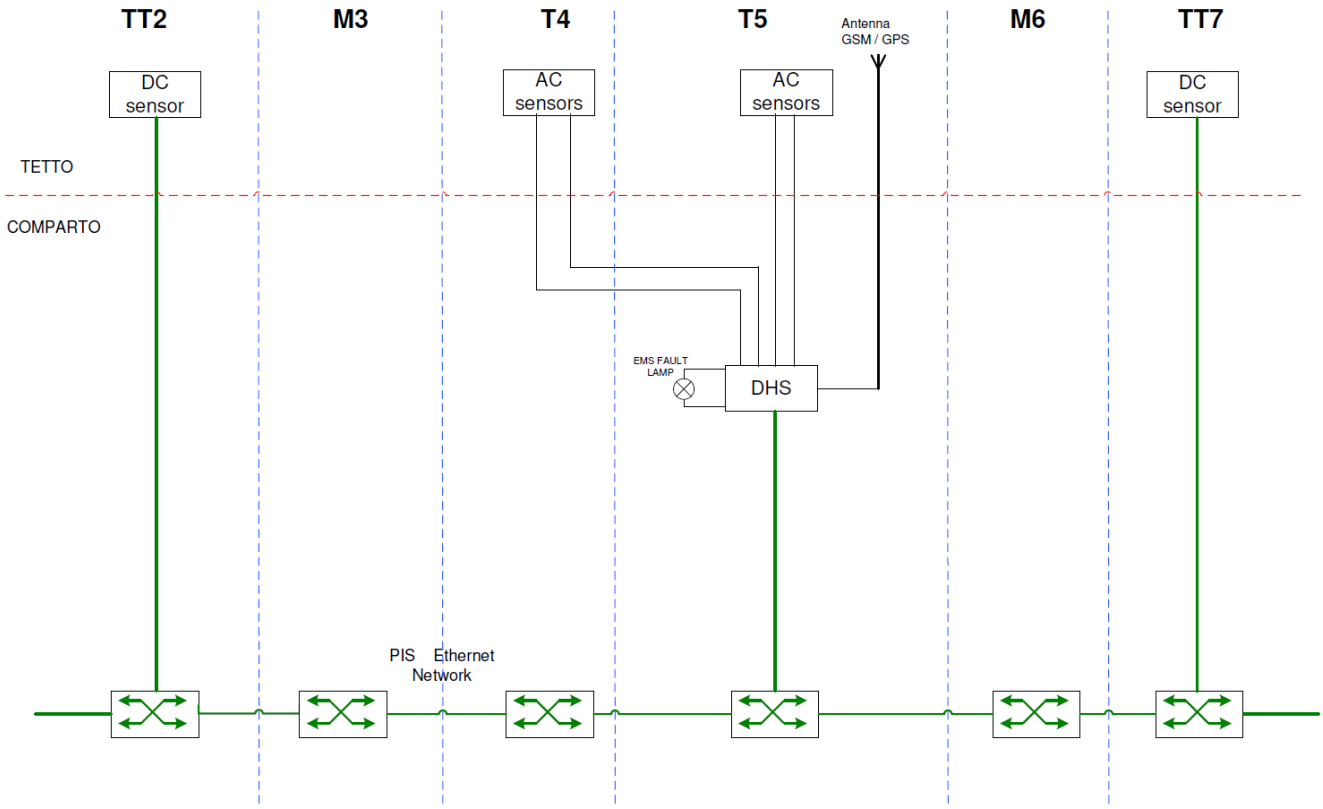


Figure 1 - EMS System

5. Requirements

HR-EMS-VZI-F-00010

The system shall be provided in accordance to the requirements and to relevant product developed in reference to the document [2.2] except from what differently specified in this document.

HR-EMS-VZI-F-00020

The system shall be compliant to TSI, EN50463-2017, and rules for cybersecurity, as indicated in document [2.1].

HR-EMS-VZI-F-00030

The DC Sensors and the DHS shall be connected to PIS Ethernet Switches in TT2, T5 and TT7 cars as indicated in Figure 1.

HR-EMS-VZI-F-00040

Dedicated VLAN for the EMS shall be configured on the PIS Network.

HR-EMS-VZI-F-00050

This VLAN shall be added to the trunk ports to enable communication between the two DC sensors and the DHS.

HR-EMS-VZI-F-00060

Relay digital output (Stall alarm) from DHS will be available for management of local diagnostic function implemented by HR vehicle logic.

HR-EMS-VZI-F-00070

The energy meter system will transmit energy stored data for billing purpose to remote DCS (both in Italy and in France) via dedicated antenna.

HR-EMS-VZI-F-00080

In addition, the energy meter system will transmit energy stored data and diagnostic data to a ground server through PIS network using Icomera router.

HR-EMS-VZI-F-00090

The supplier shall update all the system relevant documentation and manuals that are affected by the modifications requested in this document.

HR-EMS-VZI-F-00100

The Supplier shall provide HR with the test plan and related tests reports needed to validate the solution.

HR-EMS-VZI-F-00110

The Supplier shall provide HR support for certification activities required by this update.

HR-EMS-VZI-F-00120

All documentation / certification necessary for Notified Body shall be provided to HR from supplier.